# YUROU LIU

New Haven, CT · +1 (203) 808-3039 · yurou.liu@yale.edu · https://github.com/yurouninaliu ·
https://yurouninaliu.github.io · 0009-0007-9211-2884

## EDUCATION

YALE UNIVERSITY — New Haven, CT

B.S. in Physics (Intensive) — Aug 2022 - May 2026 (anticipated)
B.S. in Computer Science
GPA: 3.99/4.00 Overall; 4.00/4.00 in Physics (Intensive)
Relevant Courses: Exoplanets, Many-Body Dynamics, Observational Astronomy, Machine Learning, Classical Mechanics, Electromagnetism, Quantum Mechanics I & II, Statistical Mechanics, Algorithms, Systems Programming, Computer Graphics

## RESEARCH EXPERIENCE

INDIANA UNIVERSITY BLOOMINGTON — Bloomington, IN
Alexander P. Hixon Summer Research Fellow, Department of Astronomy — Jun 2025 - Present
*Mentored by Prof. Cristobal Petrovich and Dr. Hareesh Bhaskar*

- Developed novel secular integration codes in C to study secular chaos in hierarchical "3+1" quadruple exoplanet systems (planet orbiting a star, perturbed by a nearby substellar/stellar companion and a fourth distant stellar companion)
- Created surfaces of section from rotating frame Hamiltonian of the "3+1" quadruple system to analytically probe parameter space of chaos
- Demonstrated that near-coplanar and circular "3+1" quadruple systems can excite planetary eccentricities to levels sufficient for high-eccentricity migration

CALIFORNIA INSTITUTE OF TECHNOLOGY — Pasadena, CA
Summer Undergraduate Research Fellow, Department of Astronomy — Jun 2024 - Present
*Mentored by Dr. Yapeng Zhang and supervised by Prof. Dimitri Mawet*

- Performed atmospheric retrievals on CRIRES+ spectra of directly imaged gas giant 2M0249-0557 c and two benchmark brown dwarfs in the β Pictoris moving group
- Developed a nested sampling retrieval code using petitRADTRANS and PyMultiNest to constrain C/O ratio, metallicity, and $^{12}CO/^{13}CO$ isotopologue ratio
- Interpreted retrieved compositions in the context of formation pathways, supporting a star-like formation mechanism for 2M0249-0557 c

YALE UNIVERSITY — New Haven, CT
Undergraduate Researcher, Department of Astronomy — Jun 2023 – Present
*Mentored by Prof. Malena Rice and Dr. Tiger Lu*

- Ran N-body simulations with REBOUND and REBOUNDx to model mirrored ZLK migration of two hot Jupiters in binary star systems
- Estimated occurrence rates of double hot Jupiters using the observed distribution of Gaia binaries hosting hot Jupiters
- Quantified the impact of mass and inclination asymmetries on double hot Jupiter formation
- Results published in the Astrophysical Journal; a follow-up paper examining a case study of the WASP-94 system in preparation.

## AWARDS AND FELLOWSHIPS

Alexander P. Hixon Fellowship, Yale University, 2025
Summer Undergraduate Research Fellowship, California Institute of Technology, 2024

First-Year Summer Research Fellowship in the Sciences & Engineering, Yale University, 2023

## PUBLICATIONS

**Liu, Y.**, Lu, T. and Rice, M. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94. In Prep
**Liu, Y.**, Zhang, Y., Mawet, D., Xuan, J., Snellen, I., Stolker, T., Kesseli, A. and Rice, M. Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pictoris Young Moving Group. In Prep
**Liu, Y.**, Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems through von Zeipel–Lidov–Kozai Migration. *ApJ* 986, 103
Lu, T., Tajer, H., Hernandez, D.M., Rein, H., **Liu, Y.** and Rice, M., 2025. Collisional Fragmentation Support in TRACE. *RNAAS* 9, 110

## ORAL PRESENTATIONS (*INVITED)

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| Princeton Exoplanet Lunch* | Princeton, NJ, USA |
| *TBD* | Sep 2025 |
| Emerging Researchers in Exoplanet Science Symposium X | Princeton, NJ, USA |
| *Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group* | Jun 2025 |
| 246th American Astronomical Society Meeting | Anchorage, AK, USA |
| *Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group* | Jun 2025 |
| Boston Area Planetary Science Meeting | Boston, MA, USA |
| *Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group* | May 2025 |
| Caltech Student-Faculty Programs Seminar Day | Pasadena, CA, USA |
| *Chemistry and Isotope Ratios of the Directly Imaged Gas Giant 2MJ0249-0557c* | May 2025 |
| The 55th Meeting of the Division of Dynamical Astronomy | Toronto, Canada |
| *The Formation of Double Hot Jupiter Systems Through ZLK Migration* | May 2024 |
| Yale Astronomy Department Summer Research Symposium | New Haven, CT, USA |
| *The Dynamical History of the Two Hot Jupiters in the WASP-94 Binary Star System* | Jul, 2023 |

## POSTERS

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| New York Area Exoplanets Meeting | New York, NY, USA |
| *Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group* | May 2025 |

## OBSERVING EXPERIENCE

Palomar Observatory (DBSP): 1 night
*Spectroscopy of a Gas Giant Exoplanet in Optical Wavelength*

## TECHNICAL SKILLS

Languages: Python, C/C++, D, Rust, R
Scientific/Graphics: REBOUND, REBOUNDx, KozaiPy, petitRADTRANS, PyMultiNest, TensorFlow, PyTorch, matplotlib, NumPy, SciPy, Astropy, pandas, GSL, Mathematica, OpenGL, SDL
Tools: Git, Bash, LaTeX, SQL, Slurm

## PROJECTS

Generate Fake Transits
- Trained a conditional variational autoencoder on TESS data to generate synthetic exoplanet transits based on input parameters.

L4 and L5 Lagrange Points Visualization

- Visualized particle dynamics near Lagrange points in a star-planet system using OpenGL instanced rendering.

*More projects at:* yurouninaliu.github.io/projects

## TEACHING EXPERIENCE

YALE UNIVERSITY — New Haven, CT

Undergraduate Learning Assistant, Department of Physics — Sep 2023 – May 2024

PHYS 261: Intensive Introductory Physics II

PHYS 260: Intensive Introductory Physics I

## SCIENCE COMMUNICATION AND OUTREACH

Contributor and layout editor, Yale Scientific Magazine

- Created original artwork and designed print layouts
- Provided individualized feedback to contributors
- Improved layout designs for publication
- Led layout workshops to train new designers

Workshop leader, Yale Guild of Book Makers

- Taught hands-on workshops in bookbinding

## MEDIA

Yale News, "New study offers a double dose of 'hot Jupiters'" — Jun 2025

StarXiv, "Episode 11: Whirling planes, wandering black holes and alien supernovae" — May 2025